

Merging, Joining & Concat

In [2]: `import pandas as pd`

In [3]: `df1 = pd.read_csv(r"LOTR.csv")`
`df1`

Out[3]:

	FellowshipID	FirstName	Skills
0	1001	Frodo	Hiding
1	1002	Samwise	Gardening
2	1003	Gandalf	Spells
3	1004	Pippin	Fireworks

In [4]: `df2 = pd.read_csv(r"LOTR 2.csv")`
`df2`

Out[4]:

	FellowshipID	FirstName	Age
0	1001	Frodo	50
1	1002	Samwise	39
2	1006	Legolas	2931
3	1007	Elrond	6520
4	1008	Barromir	51

In [5]: `df1.merge(df2, how = 'inner', on = ['FellowshipID', 'FirstName'])`
#joined common (inner)

Out[5]:

	FellowshipID	FirstName	Skills	Age
0	1001	Frodo	Hiding	50
1	1002	Samwise	Gardening	39

In [6]: `df1.merge(df2, how = 'outer')` *# joined all*

Out[6]:

	FellowshipID	FirstName	Skills	Age
0	1001	Frodo	Hiding	50.0
1	1002	Samwise	Gardening	39.0
2	1003	Gandalf	Spells	NaN
3	1004	Pippin	Fireworks	NaN
4	1006	Legolas	NaN	2931.0
5	1007	Elrond	NaN	6520.0
6	1008	Barromir	NaN	51.0

In [7]: `df1.merge(df2, how = 'left') # Age column joined to left`

Out[7]:

	FellowshipID	FirstName	Skills	Age
0	1001	Frodo	Hiding	50.0
1	1002	Samwise	Gardening	39.0
2	1003	Gandalf	Spells	NaN
3	1004	Pippin	Fireworks	NaN

In [8]: `df1.merge(df2, how = 'right')`

Out[8]:

	FellowshipID	FirstName	Skills	Age
0	1001	Frodo	Hiding	50
1	1002	Samwise	Gardening	39
2	1006	Legolas	NaN	2931
3	1007	Elrond	NaN	6520
4	1008	Barromir	NaN	51

In [9]: `df1.merge(df2, how = 'cross') # Like cross product of matrix`

Out[9]:

	FellowshipID_x	FirstName_x	Skills	FellowshipID_y	FirstName_y	Age
0	1001	Frodo	Hiding	1001	Frodo	50
1	1001	Frodo	Hiding	1002	Samwise	39
2	1001	Frodo	Hiding	1006	Legolas	2931
3	1001	Frodo	Hiding	1007	Elrond	6520
4	1001	Frodo	Hiding	1008	Barromir	51
5	1002	Samwise	Gardening	1001	Frodo	50
6	1002	Samwise	Gardening	1002	Samwise	39
7	1002	Samwise	Gardening	1006	Legolas	2931
8	1002	Samwise	Gardening	1007	Elrond	6520
9	1002	Samwise	Gardening	1008	Barromir	51
10	1003	Gandalf	Spells	1001	Frodo	50
11	1003	Gandalf	Spells	1002	Samwise	39
12	1003	Gandalf	Spells	1006	Legolas	2931
13	1003	Gandalf	Spells	1007	Elrond	6520
14	1003	Gandalf	Spells	1008	Barromir	51
15	1004	Pippin	Fireworks	1001	Frodo	50
16	1004	Pippin	Fireworks	1002	Samwise	39
17	1004	Pippin	Fireworks	1006	Legolas	2931
18	1004	Pippin	Fireworks	1007	Elrond	6520
19	1004	Pippin	Fireworks	1008	Barromir	51

```
In [10]: df1.join(df2, on = 'FellowshipID', how = 'outer', lsuffix = '_Left', rsuffix = '_Rig
```

Out[10]:

	FellowshipID	FellowshipID_Left	FirstName_Left	Skills	FellowshipID_Right	Firstl
NaN	0	NaN	NaN	NaN	1001.0	
NaN	1	NaN	NaN	NaN	1002.0	
NaN	2	NaN	NaN	NaN	1006.0	
NaN	3	NaN	NaN	NaN	1007.0	
NaN	4	NaN	NaN	NaN	1008.0	
0.0	1001	1001.0	Frodo	Hiding	NaN	
1.0	1002	1002.0	Samwise	Gardening	NaN	
2.0	1003	1003.0	Gandalf	Spells	NaN	
3.0	1004	1004.0	Pippin	Fireworks	NaN	

In [11]: `df4 = df1.set_index('FellowshipID').join(df2.set_index('FellowshipID'), lsuffix = 'df4')`

Out[11]:

	FirstName_Left	Skills	FirstName_Right	Age
FellowshipID				
1001	Frodo	Hiding	Frodo	50.0
1002	Samwise	Gardening	Samwise	39.0
1003	Gandalf	Spells	NaN	NaN
1004	Pippin	Fireworks	NaN	NaN
1006	NaN	NaN	Legolas	2931.0
1007	NaN	NaN	Elrond	6520.0
1008	NaN	NaN	Barromir	51.0

In [23]: `pd.concat([df1,df2], join = 'outer', axis = 1) # concat means putting on top of ot`

Out[23]:

	FellowshipID	FirstName	Skills	FellowshipID	FirstName	Age
0	1001.0	Frodo	Hiding	1001	Frodo	50
1	1002.0	Samwise	Gardening	1002	Samwise	39
2	1003.0	Gandalf	Spells	1006	Legolas	2931
3	1004.0	Pippin	Fireworks	1007	Elrond	6520
4	NaN	NaN	NaN	1008	Barromir	51

In [24]: `pd.concat([df1, df2], ignore_index=True) # concat means putting on top of other`

Out[24]:

	FellowshipID	FirstName	Skills	Age
0	1001	Frodo	Hiding	NaN
1	1002	Samwise	Gardening	NaN
2	1003	Gandalf	Spells	NaN
3	1004	Pippin	Fireworks	NaN
4	1001	Frodo	NaN	50.0
5	1002	Samwise	NaN	39.0
6	1006	Legolas	NaN	2931.0
7	1007	Elrond	NaN	6520.0
8	1008	Barromir	NaN	51.0

```
In [21]: from IPython.display import HTML
HTML('')
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Out[21]:  No description has been provided for this image